

COMPREHENSIVE GUESS PAPER

Computer Science - Class 9 (SSC-I)

FBISE 2nd Annual Examination 2026

EXAM DATE: 25 SEPTEMBER 2026

Summative Theory SLOs Only

Based on	NBF textbook, FBISE Assessment Framework, Model Paper 2025, 1st Annual 2025 (Morning/Evening), 2nd Annual 2025, and three 1st Annual 2026 papers.
Priority system	MUST DO = revise first; HIGH = very likely concept; STRONG = completes SLO coverage.
Credit	Prepared by Fakhar STEM Sphere

Prediction notice: This is an evidence-based revision resource, not an official or leaked paper. No guess paper can guarantee exact questions or full marks; complete preparation of all summative SLOs remains essential.

1. How This Guess Was Built

The 2025-2026 papers were compared with the Model Paper, the textbook end-of-unit questions and the official Grade IX assessment framework. Repeated concepts were ranked highly, while untested but clearly summative SLOs were retained to protect students from a fresh application-style paper.

Domain	Strongest evidence	2026 focus
A. Computer Systems	CPU/ALU/CU, memory, systems, software and data communication recur throughout the papers.	Memory hierarchy, buses, topologies, transmission, OS/application software.
B. Computational Thinking	IPO, decomposition, logical/algorithmic thinking and flowcharts appear repeatedly.	Application scenarios plus decision/selection flowcharts.
C. Programming	HTML correction/output, tags and JavaScript loops/variables are consistent favourites.	DOM, CSS methods, arrays, selection/repetition and debugging.
D. Data and Analysis	Data types, collection methods and big data appear in nearly every set.	Classification with reasons, 3Vs, dataset/database, applications.
E. Applications	AI areas/applications and cloud computing are repeatedly tested.	ML types, ethical bias, education/health/agriculture, cloud models.
F. Impacts	Phishing, piracy, privacy, fake news and responsible use recur in objective and subjective parts.	Scenario-based safety, intellectual property and social/environmental impacts.
H. Entrepreneurship	Design thinking and business-plan elements are increasing in 2026 papers.	Five stages, growth mindset, SDGs, TAM/SAM/SOM and feasibility.

Official paper pattern used

Section A	12 MCQs	12 marks
Section B	10 short questions, each with OR	30 marks
Section C	3 extended questions, each with OR	18 marks

2. Core Full-Length Predicted Paper

Attempt this first under exam conditions. It follows the 2026 60-mark pattern and deliberately balances all high-frequency domains.

SECTION A - Objective (12 marks)

1. Which memory is closest to the CPU and provides very high-speed temporary storage?

- A. RAM B. Cache C. ROM D. Hard disk

2. Which bus carries the address of the memory location to be accessed?

- A. Data bus B. Control bus C. Address bus D. Expansion bus

3. Breaking a complex problem into smaller manageable parts is called:

- A. Abstraction B. Decomposition C. Debugging D. Iteration

4. Which flowchart symbol is used for a decision?

- A. Rectangle B. Parallelogram C. Diamond D. Oval

5. Which HTML tag creates a hyperlink?

- A. <link> B. <a> C. <href> D. <p>

6. Which CSS property changes text colour?

- A. font-style B. background C. color D. text-size

7. Which JavaScript structure stores several values in one variable?

- A. Array B. Boolean C. Operator D. Function call

8. Temperature is normally classified as:

- A. Nominal B. Discrete C. Continuous D. Binary

9. Which V of big data describes the speed at which data is generated and processed?

- A. Volume B. Velocity C. Variety D. Veracity

10. The AI field that enables computers to interpret images and video is:

- A. NLP B. Computer vision C. Speech recognition D. Cloud computing

11. A deceptive email asking for bank details is an example of:

- A. Freeware B. Phishing C. Backup D. Encryption

12. The first stage of design thinking is:

- A. Define B. Ideate C. Empathize D. Prototype

Answer key: 1-B, 2-C, 3-B, 4-C, 5-B, 6-C, 7-A, 8-C, 9-B, 10-B, 11-B, 12-C

SECTION B - Short Response (10 x 3 = 30)

Attempt one side of each OR pair. For full marks: define the term, give the required points, and add an example/justification where asked.

1. Explain the function of the ALU and name two operations it performs.	OR	What is problem decomposition? Explain how it helps solve a complex problem.
2. Differentiate between RAM and ROM in any three points.	OR	Differentiate between a static website and a dynamic website in any three points.
3. Why are buses important in computer architecture? State the purpose of data, address and control buses.	OR	Write HTML code to display an ordered list containing: RAM, ROM and Cache.
4. Compare half-duplex and full-duplex communication with one example of each.	OR	Explain why variables and data types are important in JavaScript, with one example.
5. For an online admission form, identify suitable inputs, main processing and outputs.	OR	Classify these as categorical or numerical and justify: blood group, number of siblings, height.
6. State three safe practices for recognizing and avoiding phishing/fake websites.	OR	Differentiate between dataset and database with a suitable example.
7. Explain how packet switching transmits data efficiently over a network.	OR	Describe Volume, Velocity and Variety of big data, using email or social media as context.
8. Correct any three errors in the code below: <code></html><head></title>My Page<title><head><body></h1>Welcome</h1><body></code>	OR	Explain one benefit and two ethical risks of using AI for human decision-making.
9. Predict and explain the output: <code>let a = 10; let b = "5"; console.log(a + b); console.log(a - b);</code>	OR	Name the five stages of design thinking and explain why the process is iterative.
10. State three functions of an operating system.	OR	Explain three benefits of cloud computing for students or a school.

SECTION C - Extended Response (3 x 6 = 18)

Q3. **[MUST DO]** Compare bus, star and ring topologies. Draw neat labelled diagrams and state at least one advantage and one limitation of each.

OR

[MUST DO] Explain AI and machine learning. Discuss the use of AI in education and healthcare, including one ethical concern.

Q4. **[MUST DO]** Draw a flowchart that inputs three unequal numbers and displays the largest. Clearly label Start/End, Input/Output, Process and Decision symbols.

OR

[MUST DO] Write a JavaScript program that inputs marks and displays grade A (90 or above), B (70-89), C (50-69) or F (below 50).

Q5. **[MUST DO]** What is big data? Explain its key characteristics and any two applications in business, healthcare, agriculture or manufacturing.

OR

[MUST DO] Explain computing innovation and discuss its beneficial and harmful impacts on users, society and the environment.

3. Comprehensive Predicted MCQ Bank

Revise all of these. Questions 1-24 are the highest-priority factual and conceptual stems; 25-56 protect the less-repeated summative SLOs.

MCQs 1-28

1. [HIGH] The technology used in first-generation computers was:
A. Transistors B. Vacuum tubes C. ICs D. Microprocessors
2. [HIGH] Integrated circuits were introduced in the:
A. First B. Second C. Third D. Fourth generation
3. [HIGH] The main purpose of the motherboard is to:
A. Print output B. Connect hardware components C. Store data permanently D. Translate programs
4. [HIGH] Which CPU unit directs and coordinates computer operations?
A. ALU B. Control Unit C. Cache D. Hard disk
5. [HIGH] Which is volatile primary memory?
A. ROM B. RAM C. DVD D. SSD
6. [HIGH] Correct order from fastest/smallest to slower/larger memory is:
A. Registers, cache, RAM, secondary storage B. RAM, registers, cache, HDD C. HDD, RAM, cache, registers D. Cache, registers, HDD, RAM
7. [HIGH] Which software manages hardware and provides services to applications?
A. Application software B. System software C. Presentation software D. Spreadsheet
8. [HIGH] A word processor is an example of:
A. System software B. Application software C. Firmware D. Low-level language
9. [HIGH] In simplex mode, data travels:
A. Both ways simultaneously B. Both ways but one at a time C. One way only D. Without a medium
10. [HIGH] Which topology connects every device to a central device?
A. Bus B. Ring C. Star D. Mesh
11. [HIGH] A router mainly:
A. Prints documents B. Connects different networks C. Stores webpages D. Performs arithmetic
12. [HIGH] Which transmission method sends data in blocks with shared timing?
A. Asynchronous B. Synchronous C. Simplex D. Half-duplex
13. [HIGH] The correct IPO order is:
A. Input-Program-Output B. Input-Process-Output C. Input-Plan-Output D. Input-Proceed-Output
14. [HIGH] A yes/no computing problem is a:
A. Counting problem B. Search problem C. Decision problem D. Design problem
15. [HIGH] Logical thinking mainly helps by:
A. Arranging reasoning in a valid sequence B. Increasing storage C. Designing graphics D. Replacing algorithms
16. [HIGH] A rectangle in a flowchart represents:
A. Decision B. Input/output C. Process D. Start/end
17. [HIGH] Which is NOT a standard design method named in the framework?
A. Flowchart B. Concept map C. IPO chart D. Random guessing
18. [HIGH] A website mainly provides information, while a web application:
A. Cannot use a browser B. Performs interactive tasks C. Has no server D. Contains only images
19. [HIGH] The top node in an HTML DOM tree is:
A. body B. head C. document D. title
20. [HIGH] Which tag gives the main heading?
A. <h1> B. <p> C. <title> D. <head>
21. [HIGH] Which attribute stores the destination of an anchor link?
A. src B. href C. style D. name
22. [HIGH] Which CSS method is stored in a separate .css file?
A. Inline B. Internal C. External D. Embedded JavaScript
23. [HIGH] Which keyword declares a block-scoped JavaScript variable?
A. dim B. let C. define D. integer
24. [HIGH] Which operator gives the remainder in JavaScript?
A. / B. % C. * D. +

25. [STRONG] Which statement is used for selection?

A. if...else B. for C. array D. alert

26. [STRONG] A breakpoint is used to:

A. Stop execution temporarily for inspection B. Delete code C. Publish a site D. Create HTML tags

27. [STRONG] A collection of related values stored under one JavaScript name is an:

A. Operator B. Array C. Comment D. Tag

28. [STRONG] The output of `let x=5; x=x+2;` is:

A. 5 B. 7 C. 52 D. Error

MCQs 29-56

29. [STRONG] Blood group is which type of attribute?

A. Nominal B. Ordinal C. Discrete D. Continuous

30. [STRONG] A satisfaction rating low/medium/high is:

A. Nominal B. Ordinal C. Continuous D. Ratio-scaled

31. [STRONG] Number of students in a class is:

A. Continuous B. Discrete C. Nominal D. Binary

32. [STRONG] A collection of related data in context is a:

A. Protocol B. Dataset C. Topology D. Function

33. [STRONG] Interviews and observations are commonly:

A. Primary data methods B. Secondary storage C. Network modes D. CSS methods

34. [STRONG] Government records used by a researcher are usually:

A. Primary data B. Secondary data C. Continuous data D. Binary data

35. [STRONG] Which big-data characteristic refers to many forms such as text, image and video?

A. Volume B. Velocity C. Variety D. Value

36. [STRONG] Data analytics primarily examines raw data to:

A. Install hardware B. Draw conclusions C. Create passwords D. Route packets

37. [STRONG] Learning from labelled input-output examples is:

A. Unsupervised learning B. Supervised learning C. Cloud computing D. Robotics

38. [STRONG] Finding hidden groups in unlabelled data is:

A. Supervised learning B. Unsupervised learning C. Packet switching D. Encryption

39. [STRONG] Understanding human language is the goal of:

A. NLP B. Computer vision C. Robotics D. CSS

40. [STRONG] Siri is an example of a:

A. Compiler B. Virtual assistant C. Network topology D. Spreadsheet

41. [STRONG] A cloud owned and used by one organisation is:

A. Public cloud B. Private cloud C. Hybrid cloud D. Remote cloud

42. [STRONG] An AI decision that unfairly disadvantages a group demonstrates:

A. Compression B. Bias C. Encapsulation D. Scalability

43. [STRONG] Using cracked commercial software is:

A. Fair use B. Software piracy C. Freeware D. Open source

44. [STRONG] Software that secretly monitors a user is:

A. Shareware B. Spyware C. Firmware D. Freeware

45. [STRONG] Redirecting a user to a fake website is commonly called:

A. Pharming B. Backup C. Debugging D. Prototyping

46. [STRONG] A legal right protecting an original software work is:

A. Copyright B. Protocol C. Cookie D. Topology

47. [STRONG] Which is the safest password?

A. pakistan123 B. Fakhar2000 C. T9!qL2#vP7 D. 12345678

48. [STRONG] A reliable online source should be checked for:

A. Only colourful design B. Author, evidence, date and corroboration C. Most shares only D. Anonymous claims

49. [STRONG] Electronic waste is primarily a(n):

A. Environmental impact B. HTML error C. Data type D. Communication mode

50. [STRONG] The design-thinking stage that studies users' needs is:

A. Empathize B. Ideate C. Prototype D. Test

51. [STRONG] Generating many possible solutions occurs in:

A. Define B. Ideate C. Test D. Finance

52. [STRONG] A sample model of a solution is a:

A. Protocol B. Prototype C. Dataset D. Patent

53. [STRONG] A growth mindset assumes abilities can:

A. Never change B. Improve through effort and learning C. Only be inherited D. Replace market research

54. [STRONG] TAM in a business plan means:

A. Total Addressable Market B. Target Advertising Method C. Total Annual Money D. Technical Access Model

55. [STRONG] A unique selling proposition explains:

A. Why the offering is distinct and valuable B. Only the owner's name C. Computer memory size D. A network address

56. [STRONG] Financial feasibility asks whether the idea:

A. Can be technically typed B. Is economically workable C. Uses a star topology D. Has HTML comments

MCQ Answer Key

1. B	2. C	3. B	4. B	5. B	6. A	7. B	8. B
9. C	10. C	11. B	12. B	13. B	14. C	15. A	16. C
17. D	18. B	19. C	20. A	21. B	22. C	23. B	24. B
25. A	26. A	27. B	28. B	29. A	30. B	31. B	32. B
33. A	34. B	35. C	36. B	37. B	38. B	39. A	40. B
41. B	42. B	43. B	44. B	45. A	46. A	47. C	48. B
49. A	50. A	51. B	52. B	53. B	54. A	55. A	56. B

Common traps: RAM is volatile but ROM is non-volatile; '<title>' names the browser tab while '<h1>' displays a main heading; CSS styles a page while JavaScript adds behaviour; continuous data may take any value in an interval, whereas discrete data is countable.

4. Comprehensive Short-Question Bank

Prepare a 3-mark response for each: one clear definition/claim, two or three accurate points, and an example or reason where appropriate.

Unit 1 - Computer Systems

1. [MUST DO] Differentiate natural and artificial systems with examples.
2. [HIGH] Draw and explain the basic Von Neumann architecture.
3. [MUST DO] State the functions of CPU, ALU, CU and registers.
4. [MUST DO] Differentiate primary, cache and secondary memory.
5. [HIGH] Explain memory hierarchy in terms of speed, capacity, cost and volatility.
6. [MUST DO] Differentiate system software and application software with examples.
7. [MUST DO] State three main functions of an operating system.
8. [MUST DO] Explain data, address and control buses.
9. [HIGH] Differentiate synchronous and asynchronous transmission.
10. [MUST DO] Differentiate simplex, half-duplex and full-duplex modes with examples.
11. [HIGH] State the roles of hub, switch, router and gateway.
12. [HIGH] Give three advantages and three disadvantages of wireless networks.

Unit 2 - Computational Thinking and Algorithms

1. [MUST DO] Differentiate simple and complex problems with examples.
2. [MUST DO] Explain logical thinking and algorithmic thinking.
3. [MUST DO] What is decomposition and why is it useful?
4. [HIGH] Explain the four steps for solving a simple problem.
5. [MUST DO] Identify input, process and output for calculating average marks.
6. [MUST DO] State the importance of IPO charts and flowcharts.
7. [HIGH] Differentiate decision, search and counting problems with examples.

Unit 3 - Programming Fundamentals

1. [HIGH] Differentiate website and web application.
2. [HIGH] Differentiate front-end and back-end development.
3. [MUST DO] Differentiate static and dynamic websites.
4. [HIGH] Explain the DOM tree of an HTML document.
5. [MUST DO] State the purposes of html, head, title, body, h1, p, a, ol and ul tags.
6. [HIGH] Differentiate inline, internal and external CSS.
7. [MUST DO] Explain JavaScript variables, data types and arrays with examples.
8. [MUST DO] Explain sequence, selection and repetition.
9. [HIGH] Why is debugging important? Explain errors, testing and breakpoints.

Unit 4 - Data and Analysis

1. [MUST DO] Differentiate data science and data analytics.
2. [MUST DO] Differentiate data and dataset; also distinguish dataset and database.
3. [MUST DO] Differentiate primary and secondary data collection with examples.
4. [HIGH] Differentiate qualitative and quantitative data.
5. [HIGH] Differentiate nominal, binary and ordinal attributes.

6. [MUST DO] Differentiate discrete and continuous data with examples.
7. [MUST DO] Explain Volume, Velocity and Variety of big data.
8. [HIGH] State three benefits and two challenges of big data.

Units 5, 6 and 7

1. [MUST DO] Differentiate AI and machine learning.
2. [HIGH] Differentiate supervised and unsupervised machine learning.
3. [MUST DO] Explain any three areas of AI.
4. [MUST DO] State three benefits of cloud computing.
5. [HIGH] Explain one bias/privacy issue in AI decision-making.
6. [MUST DO] Differentiate phishing, pharming, spam, spyware and malware.
7. [HIGH] Explain software piracy, freeware, shareware and open-source software.
8. [MUST DO] How can students identify fake news and credible sources?
9. [HIGH] Explain privacy, system security and usability trade-offs.
10. [HIGH] Explain copyrights, patents and trademarks.
11. [MUST DO] List and explain the five stages of design thinking.
12. [HIGH] Explain growth mindset and its value for entrepreneurs.
13. [HIGH] State key parts of a business plan: market, USP, feasibility and go-to-market strategy.

5. Comprehensive Long/Application Question Bank

These are the most plausible 6-mark questions. Practise diagrams, code and scenario-based justification; do not memorize only definitions.

1. **[STRONG]** Explain the evolution of first- to fifth-generation computers, identifying the key technology and major improvement in each generation.
2. **[HIGH]** Explain Von Neumann architecture and show how CPU, memory, input/output and buses work together to execute instructions.
3. **[MUST DO]** Compare bus, star and ring topologies with labelled diagrams, advantages, disadvantages and a suitable use of each.
4. **[HIGH]** Explain data communication components and compare synchronous/asynchronous and simplex/half-duplex/full-duplex transmission.
5. **[MUST DO]** Explain computational thinking and its principles. Show how decomposition, logical thinking, IPO and algorithms solve a real-life problem.
6. **[MUST DO]** Draw a flowchart to determine whether an entered number is positive, negative or zero.
7. **[MUST DO]** Draw a flowchart to find the largest of three numbers.
8. **[HIGH]** Draw a flowchart to input five numbers and display their sum and average.
9. **[MUST DO]** Write a complete HTML page containing title, main heading, paragraph, hyperlink, ordered list and an image.
10. **[HIGH]** Explain three ways of applying CSS. Write HTML/CSS code that changes text colour, font size, background colour and table border.
11. **[MUST DO]** Write JavaScript using if...else to assign grades from marks; explain the variables, operators and selection used.
12. **[MUST DO]** Write JavaScript loops to (a) display 1 to 20, (b) display 10 to 1, and (c) print even numbers from 2 to 20.
13. **[HIGH]** Explain JavaScript debugging and correct a supplied code fragment containing syntax, variable-name and missing-quotation errors.
14. **[HIGH]** Explain data science and major data types with examples. Classify at least six school-data attributes and justify each classification.
15. **[MUST DO]** Define big data, explain its characteristics, benefits and challenges, and discuss applications in healthcare and manufacturing.
16. **[MUST DO]** Define AI and machine learning; explain speech recognition, computer vision, NLP and expert systems with applications.
17. **[MUST DO]** Discuss applications of AI in education, healthcare, agriculture and gaming, including bias, privacy, errors and job-displacement risks.
18. **[HIGH]** Explain safe and responsible computer use: updates, strong passwords, backups, secure Wi-Fi, privacy settings and safe downloads.
19. **[MUST DO]** Explain computing innovation and evaluate its personal, ethical, social, economic, cultural and environmental impacts.
20. **[HIGH]** Apply the five stages of design thinking to a local problem linked to an SDG, then outline a business plan including TAM/SAM/SOM, USP, financial feasibility and go-to-market strategy.

6. Extra High-Value Practice: HTML, CSS, JavaScript and Flowcharts

These questions directly address the requested extra practice and the application level emphasized in the assessment framework.

1. **[HIGH]** HTML correction: Correct all errors and rewrite a valid page:

```
<html></head><title>School<title></head></body><h2>Welcome</h1><p>CS
Class<p><body></html>
```
2. **[STRONG]** HTML output: Draw/write the browser output of code containing `<h1>`, `<p>`, `<b>`, `<i>`, `<u>`, `<ol>` and `<a>` tags.
3. **[STRONG]** HTML table: Create a 3-column table for Name, Roll No. and Marks, with a heading row and two student records.
4. **[STRONG]** HTML media: Write code to add an image with alternate text and a video that displays controls.
5. **[HIGH]** CSS selectors: Use an internal style block to make all paragraphs blue and 18px, and assign a special class with a yellow background.
6. **[STRONG]** External CSS: Show the HTML `<link>` tag and a separate CSS rule used to style h1 headings.
7. **[HIGH]** JavaScript types: Predict and explain: `let x=8; let y="2"; console.log(x+y); console.log(x*y);`
8. **[HIGH]** JavaScript array: Create an array of five subjects, use a loop to display every value, and state the array length.
9. **[HIGH]** JavaScript selection: Input a number and display Even or Odd using the remainder operator.
10. **[STRONG]** JavaScript nested selection: Input three numbers and display the greatest; also handle equal values.
11. **[HIGH]** JavaScript repetition: Use a for loop to display the multiplication table of a user-supplied number.
12. **[HIGH]** Debugging: Correct: `let fruits = ["Apple", "Banana", "Orange"]; console.log(fruit.lenght)` and identify each error.
13. **[STRONG]** Flowchart 1: Input length and width; display area and perimeter of a rectangle.
14. **[HIGH]** Flowchart 2: Input a number; display Positive, Negative or Zero.
15. **[HIGH]** Flowchart 3: Input marks in five subjects; display total, percentage and Pass/Fail using a stated pass rule.
16. **[STRONG]** Flowchart 4: Input N; use repetition to display the sum of integers from 1 to N.

Flowchart symbol checklist

Symbol	Meaning	Use
Oval / terminator	Start or End	Exactly one clear entry and exit where appropriate.
Parallelogram	Input / Output	Read values or display a result.
Rectangle	Process	Calculation or assignment.
Diamond	Decision	Label outgoing branches Yes/No or True/False.
Arrow	Flow line	Shows correct sequence and loop-back direction.

7. Final 7-Day Revision Order

Order	Coverage
Day 1	CPU, ALU/CU/registers, memory hierarchy, software and OS
Day 2	Data communication, buses, transmission modes and topologies
Day 3	Computational thinking, IPO and four flowcharts
Day 4	HTML structure/tags, DOM, CSS and error correction
Day 5	JavaScript variables, types, arrays, if...else, loops and debugging
Day 6	Data types, collection, big data, AI/ML and cloud
Day 7	Safety/ethics, computing impacts, design thinking, business plan; attempt the core paper

Full-marks response checklist

- Answer exactly what is asked and match the number of required points.
- For differences, use a two-column comparison and include examples.
- For 6-mark answers, use headings, accurate terms, examples and a labelled diagram/code where relevant.
- Trace JavaScript code line by line; mention data types when explaining unexpected output.
- Use correct flowchart shapes, arrows and labelled decision branches.
- Attempt both sides only during practice; in the exam follow the paper instructions.

Prepared by Fakhar STEM Sphere

For educational revision. Students should also revise the complete official summative theory SLOs and their textbook examples.